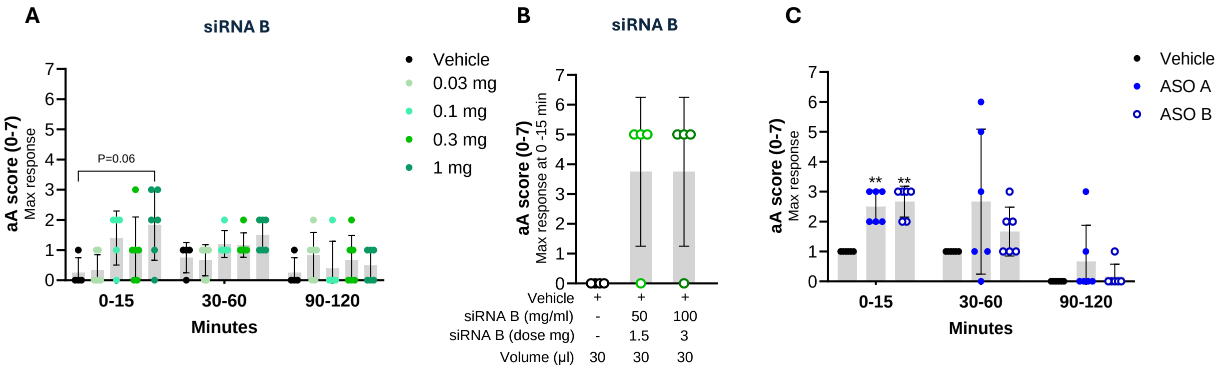
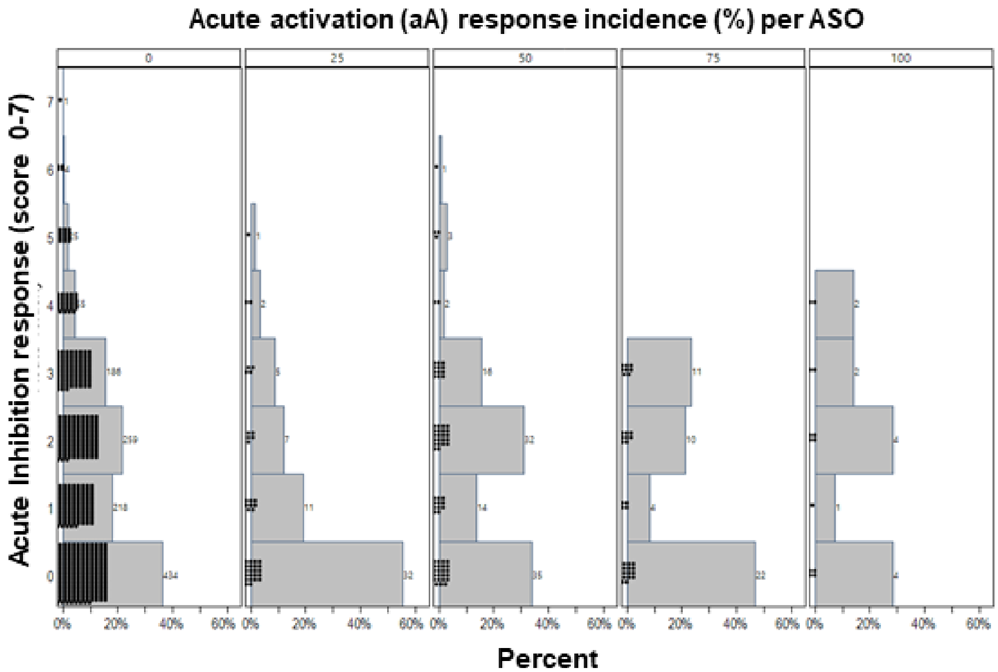
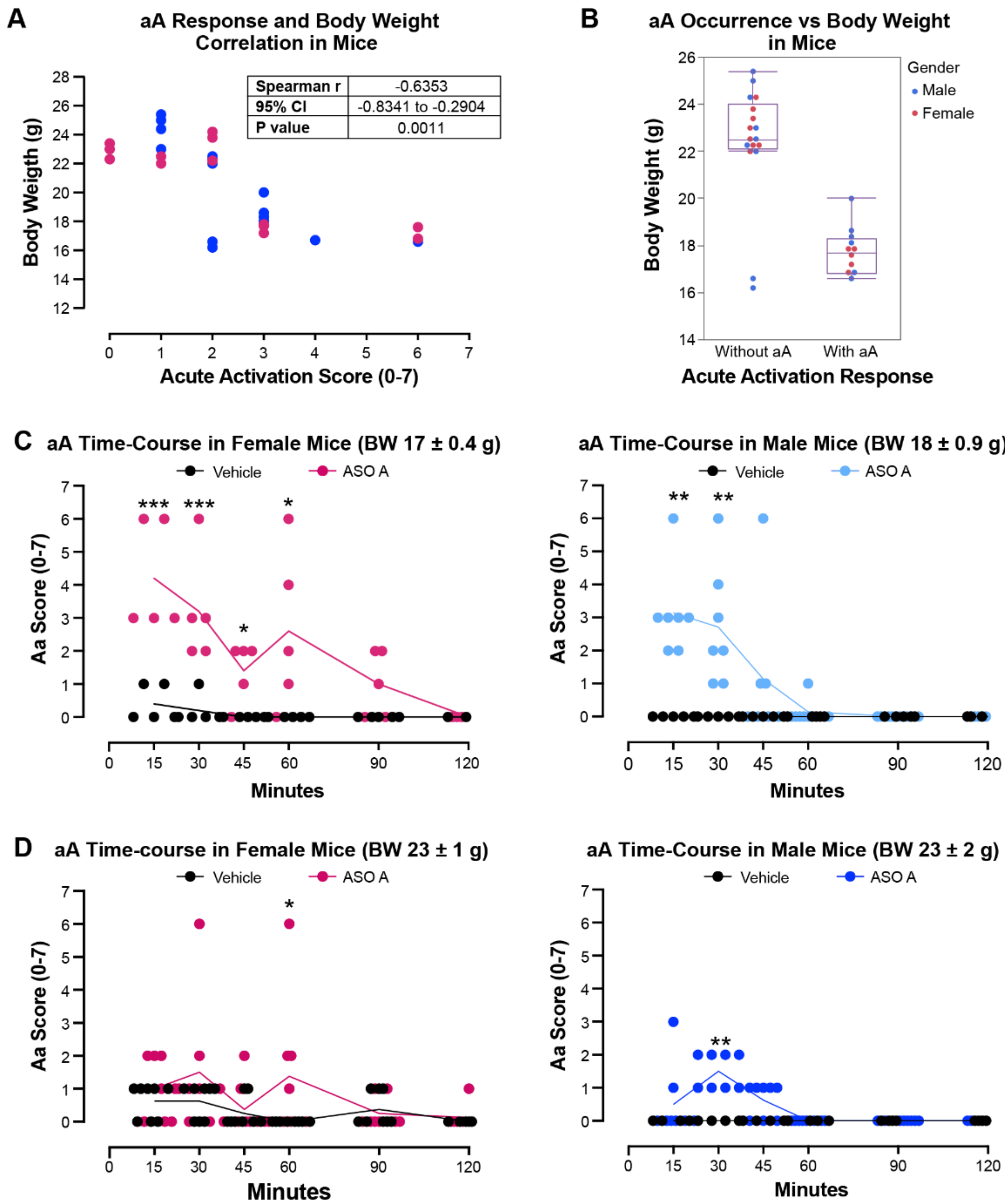


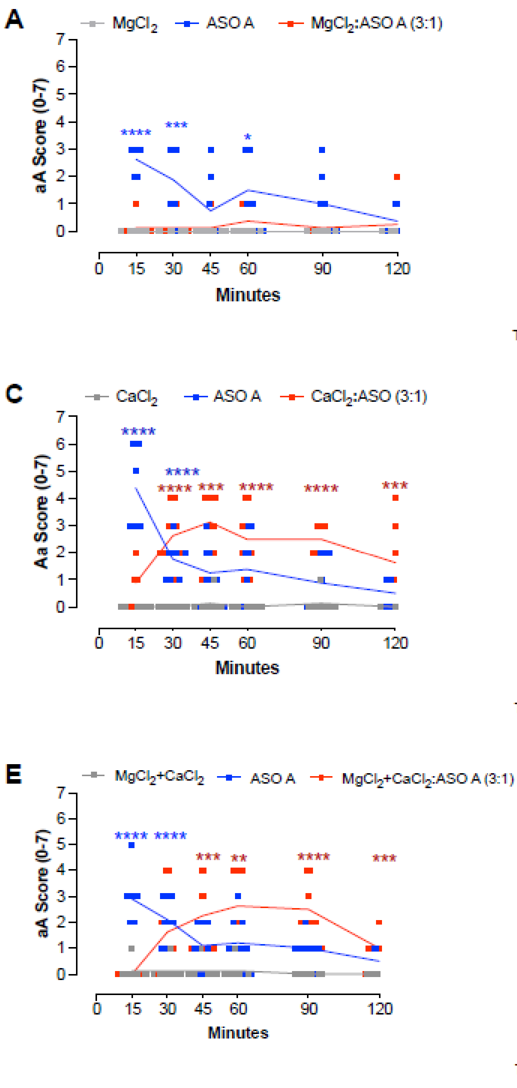
Supplementary Figure 1. Acute neuronal activation response after intrathecal administration of oligonucleotides in male rats. (A) Time-course of the acute neuronal activation (aA) response in male rats (body weight = 286 ± 9 g, n=4-6) after IT administration of siRNA B at doses of 0.03, 0.1, 0.3, 1 mg in 30 µl or 30 µl of artificial cerebrospinal fluid (Vehicle). Data are expressed as mean ± s.d. The aA scores for each group were divided into three time blocks (0-15 minutes, 30-60 minutes, and 90-120 minutes). Statistical significance was determined by the Mann-Whitney test vs vehicle (aCSF) at each dose and time point. **(B)** Acute neuronal activation response after 0-15 min of a single IT administration of 1.5 g, or 3 mg of siRNA B or PBS (Vehicle) with body weight = 384 ± 32 g (n=4). Data are expressed as mean ± s.d. The statistical significance was determined by the Kruskal-Wallis test vs vehicle (PBS). **(C)** Acute neuronal activation response after 0-15 min of a single IT administration of ASO A (blue close circles) and ASO B (blue open circles) at 3 mg in 30 µl or 30 µl of artificial cerebrospinal fluid (Vehicle) with body weight = 285 ± 4 g (n=6). Data are expressed as mean ± s.d. The aA scores for each group were divided into three time blocks (0-15 minutes, 30-60 minutes, and 90-120 minutes). Statistical significance was determined by the Mann-Whitney test vs vehicle (aCSF). No statistical significance was found between ASO A and ASO B. *P ≤ 0.05; **P ≤ 0.01; ***P ≤ 0.001; ****P ≤ 0.0001 at each dose and time point. The specific number of animals per group, body weight, and P value are included in the supplementary table 4-6. Abbreviations: IT=intrathecal; aA=Acute neuronal activation; aCSF: artificial cerebrospinal fluid.



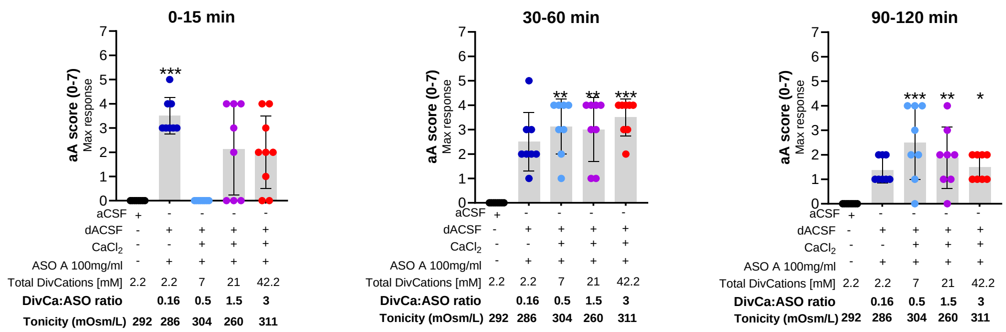
Supplementary Figure 2. The acute inhibition response does not have an associated increased incidence of acute neuronal activation response in rats. Comparing the severity of acute inhibition using the Acute Inhibition scale (score 0-7) at 3 h after ASO administration (3 mg dose) to the incidence of acute neuronal activation (aA) response. Each panel represents a level of aA incidence from 0 to 100% in 25% increments. The histogram within each panel is expressed as the percentage of ASO with a given level of acute inhibition score (average score per ASO). Each dot represents a unique ASO tested in independent studies with n≥ 4 rats in each experimental group. The figure suggests there is no connection between the occurrence of acute inhibition and the presence of acute neuronal activation response.



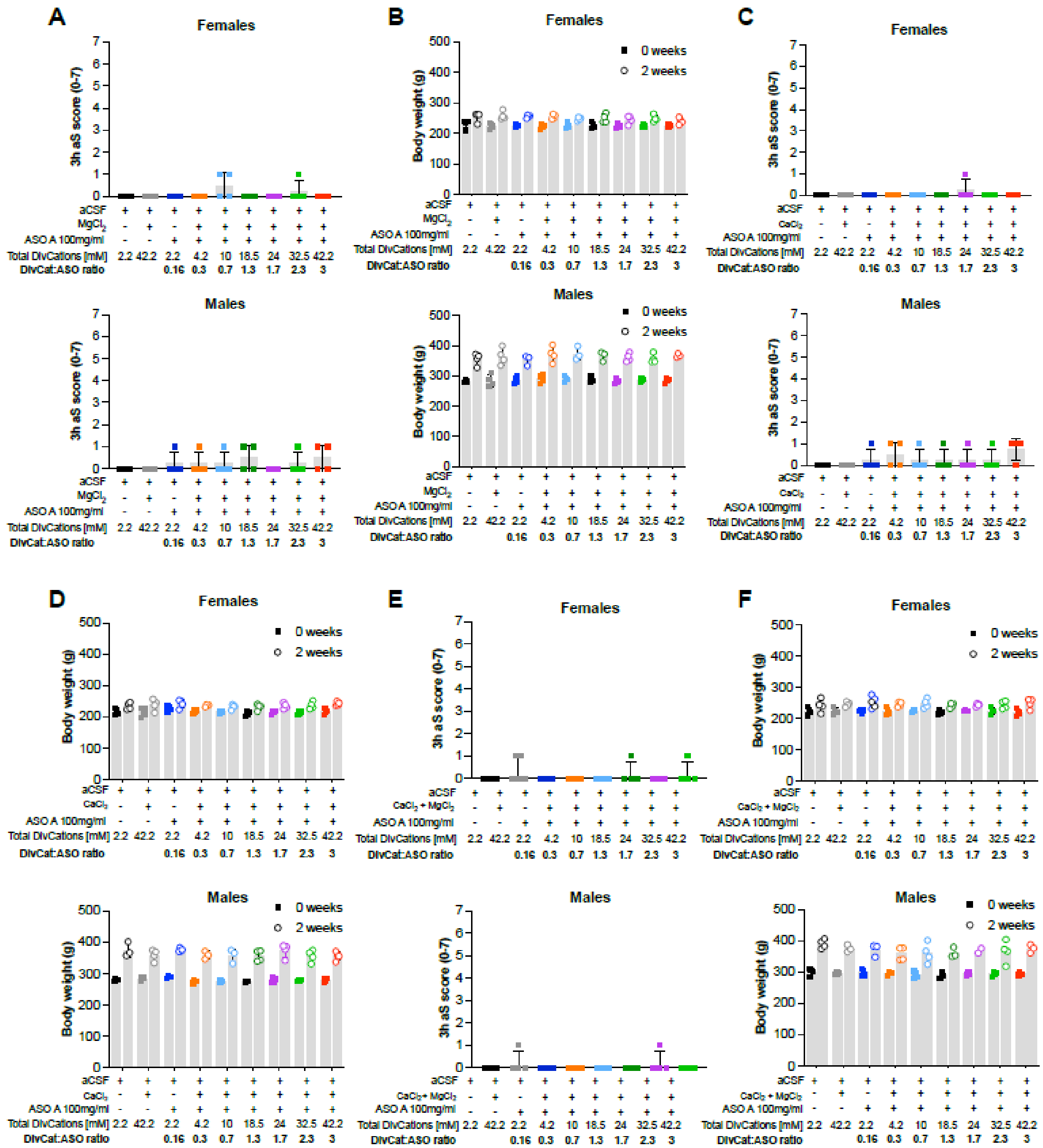
Supplementary Figure 3. Body-weight dependent, transient acute neuronal activation response after oligonucleotide injection in mice. (A) Correlation graph of acute neuronal activation score versus mouse body weight; each dot represents a unique animal. Note how animals with lower body weight had a higher aA score (females, pink circles; males, blue circles). **(B)** Occurrence of acute neuronal activation response and the mice's body weight; each dot represents a unique animal. Mice were more likely to experience at least one acute neuronal activation event (score ≥ 3 within the first 15 minutes) when body weights were lower than 20 g. All mice received a single 10 µl ICV injection of 1000 µg of ASO A. **(C-D)** Time-course of aA score after single intracerebroventricular (ICV) administration of 1000 µg ASO A in 10 µl or 10 µl of vehicle in females and males with BW ~17-18 g **(C)**; or females and males with BW ~23 g **(D)**. Each symbol represents an individual animal. The specific number of animals per group and body weight are included in the supplementary tables 10-11. The data is expressed as the mean ± s.d. The statistical significance was determined as follows: Spearman r correlation and linear regression (A); parametric unpaired t-test (B) Multiple Mann-Whitney test (C-D). *P ≤ 0.05; ** P ≤ 0.01 *** P ≤ 0.001; **** P ≤ 0.0001 *vs* vehicle (aCSF). Abbreviations: ICV=intracerebroventricular; aA=Acute neuronal activation; aCSF=artificial cerebrospinal fluid.



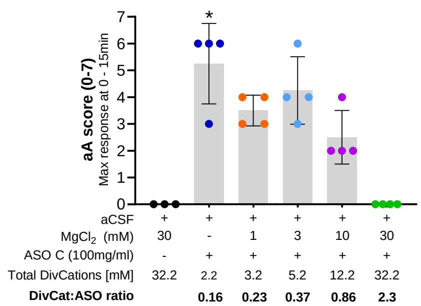
Supplementary Figure 4. Time course of ASO-induced acute neuronal activation response modifying the divalent cation-to-oligo antisense ratio with magnesium or calcium supplementation or the combination of both in rats. Time course (**A, C, E**) and area under the curve (**B, D, F**; females in pink, males in blue) of acute neuronal activation (aA) response in rats. **(A-B)** aA response after intrathecal (IT) 3 mg dose administration of ASO A formulated in aCSF with or without increasing millimolar concentrations of magnesium supplementation, resulting in six different DivCat: ASO ratios from 0.3:1 to 3:1. **(C-D)** aA response in rats after IT 3 mg dose administration of ASO A formulated in aCSF with or without increasing millimolar concentrations of calcium supplementation, resulting in six different DivCat: ASO ratios from 0.3:1 to 3:1. **(E-F)** aA response in rats after IT 3 mg dose administration of ASO A formulated in aCSF with or without increasing millimolar concentrations of magnesium plus calcium supplementation (kept at their physiological ratio), resulting in six different DivCat: ASO ratios from 0.3:1 to 3:1. Each symbol represents an individual animal. Data are expressed as the time course of the aA response (A, C, E) or the area under the curve (mean aA score within 120 min) (B, D, F). The statistical significance was determined by the Multiple Mann-Whitney test (A, C, E), and the area under the curve (AUC) was calculated using the trapezoid rule (B, D, F). The specific number of animals per group and body weight are included in the corresponding supplementary tables 12-14. P ≤ 0.05; ** P ≤ 0.01 *** P ≤ 0.001; **** P ≤ 0.0001 *vs* aCSF + 42.2 mM magnesium, calcium, or magnesium + calcium, respectively. Abbreviations: aA= Acute neuronal activation; AUC= area under the curve; aCSF = artificial cerebrospinal fluid.



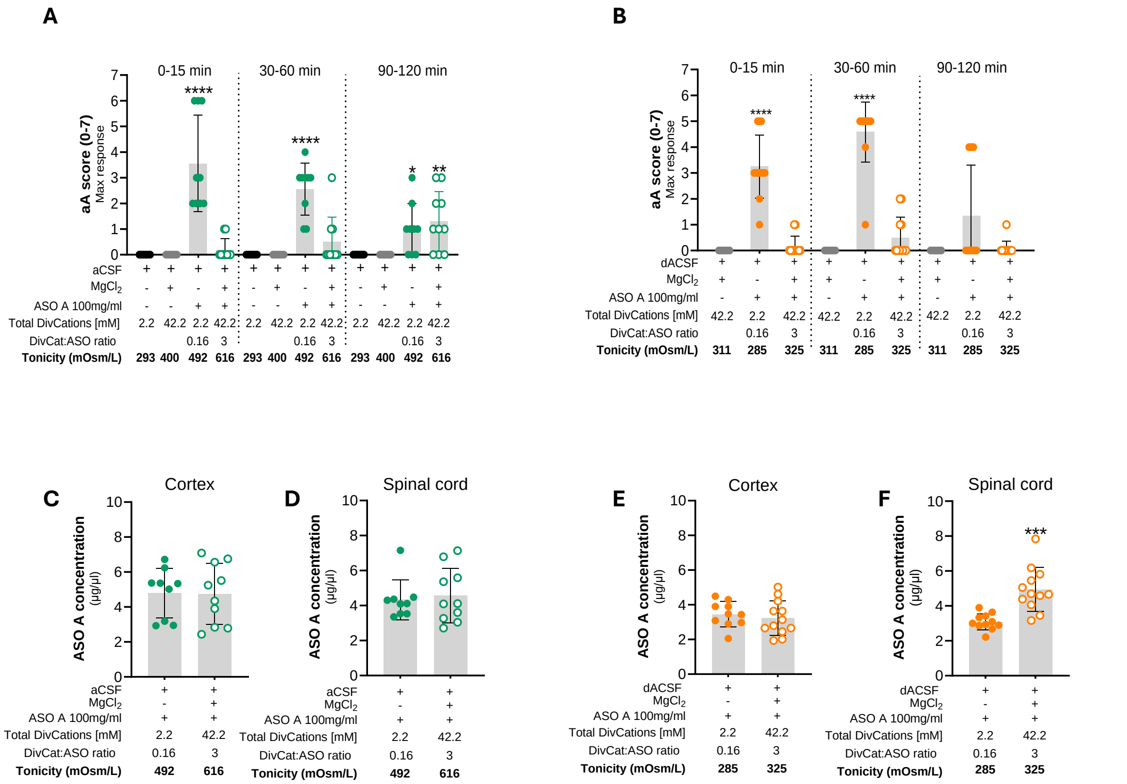
Supplementary Figure 5. Effect of increasing the divalent cation-to-oligo antisense ratio by calcium supplementation under isotonic conditions in rats. Acute neuronal activation (aA) maximum response plotted in three-time blocks; 0-15 minutes (left); 30-60 minutes (middle); and 90-120 minutes (right) after intrathecal 3 mg dose administration of ASO A formulated in the sodium-deficient artificial cerebrospinal fluid (dACSF) with or without increasing millimolar concentrations of calcium supplementation, resulting in six different DivCat: ASO ratio from 0.5:1 to 3:1 (BW= 225 ± 2 g and males, BW= 300 ± 5 g, n≥4). The data shows a significant reduction of aA response with a 0.5:1 DivCat: ASO, but not 1.5:1 or 3:1 at 0-15 min time point. The dACSF was not tested alone as tonicity is under physiological levels. Each symbol represents an individual animal. The data is expressed as the mean ± s.d. The specific number of animals per group and P value are included in the supplementary table 15. The statistical significance was determined by the Kruskal-W allis test. *P ≤ 0.05; ** P ≤ 0.01 *** P ≤ 0.001; **** P ≤ 0.0001 *vs* vehicle (aCSF). Abbreviations: aA= Acute neuronal activation; aCSF=artificial cerebrospinal fluid; dACSF= sodium deficient artificial cerebrospinal fluid (isotonic).



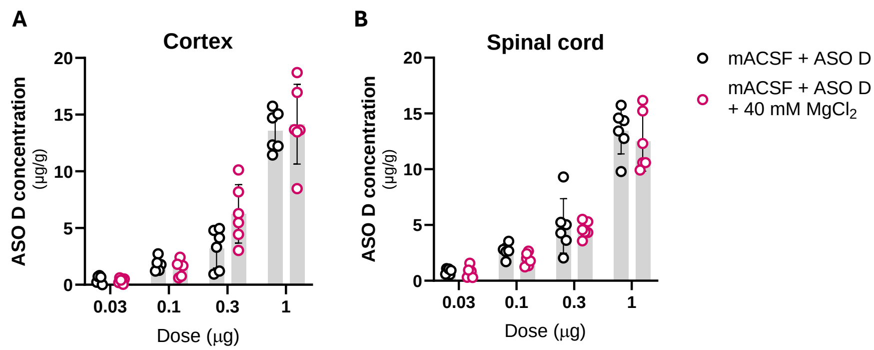
Supplementary Figure 6. Modifying the divalent cation-to-oligo antisense ratio with magnesium or calcium supplementation or the combination of both do not induce acute inhibition phenotype, nor affect body weight gain in rats. Acute inhibition response 3 hours (A, C, E) and body weight (BW) gain 2 weeks (B, D, F) after intrathecal administration of 3 mg/ 30 µl of ASO A in female (top) and male (bottom) rats. **(A-B)** Acute inhibition and BW gain after IT ASO with or without varying millimolar concentrations of magnesium supplementation formulated in aCSF, resulting in six different DivCat: ASO ratios, ranging from 0.3:1 to 3:1 (females and males, n≥ 4). **(C-D)** Acute inhibition and BW gain after IT ASO with or without varying millimolar concentrations of calcium supplementation formulated in aCSF, resulting in six different DivCat: ASO ratios, ranging from 0.3:1 to 3:1 (females and males, n≥ 4). **(E-F)** Acute inhibition and BW gain after IT ASO with or without varying millimolar concentrations of magnesium plus calcium supplementation (kept at their physiological ratio) formulated in aCSF, resulting in six different DivCat: ASO ratios, ranging from 0.3:1 to 3:1 (females and males, n≥ 4). Body weight gain was not affected in any experimental group, and no relevant acute inhibition response was observed. Each point represents an individual animal. The specific number of animals per group and body weight are included in the corresponding supplementary tables 12-14. Data are expressed as mean ± s.d. The statistical significance was determined by the Kruskal-Wallis test. *P ≤ 0.05; ** P ≤ 0.01 *** P ≤ 0.001; **** P ≤ 0.0001 *vs* vehicle (aCSF). Abbreviations: aCSF; artificial cerebrospinal fluid.



Supplementary Figure 7. ASO C-induced-acute neuronal activation is mitigated by modifying the divalent cation-to-oligo antisense ratio with magnesium supplementation in male rats. Acute neuronal activation maximum response 0-15 min after intrathecal 3 mg dose administration of ASO C with or without increasing millimolar concentrations of magnesium supplementation formulated in aCSF, resulting in four different DivCat: ASO ratios from 0.23:1 to 2.3:1 (BW = 263 ± 8 g, n=4). Each point represents an individual animal. Data are expressed as mean ± s.d. The specific number of animals per group, body weight, and P value are included in the supplementary table 16. The statistical significance was determined by the Kruskal-Wallis test. *P ≤ 0.05; ** P ≤ 0.01 *** P ≤ 0.001; **** P ≤ 0.0001 vs vehicle (aCSF). Abbreviations: aCSF, artificial cerebrospinal fluid; aA, Acute neuronal activation.



Supplementary Figure 8. Modifying the divalent cation-to-oligo antisense ratio with magnesium supplementation mitigates ASO-induced acute neuronal activation response under hypertonic or isotonic conditions in rats. (A-B) The acute neuronal activation maximum response was plotted in three time blocks: 0-15 minutes (left); 30-60 minutes (middle); and 90-120 minutes (right). **(A)** aA response after intrathecal (IT) 3 mg dose administration of ASO A with or without 40 mM of magnesium supplementation for a 3:1 DivCat: ASO ratio formulated in artificial cerebrospinal fluid (aCSF) in rats (females, BW= 204 ± 4 g and males, BW= 261 ± 5 g; n≥4). **(B)** aA response after IT 3 mg dose administration of ASO A with or without 40 mM of magnesium supplementation for a 3:1 DivCat: ASO ratio formulated in tonicity controlled (isotonic) artificial cerebrospinal fluid (sodium deficient aCSF, dACSF) in rats (females, BW= 201 ± 3 g and males, BW= 271 ± 2 g; n≥4). The specific number of animals per group, body weight, and P value are included in the supplementary table 17. **(C-D)** ASO tissue accumulation in the cortex (**C**) or spinal cord (**D**) of rats treated with an IT 3 mg dose of ASO A with or without magnesium supplementation in standard aCSF. **(E-F)** ASO tissue accumulation in the cortex (**E**) or spinal cord (**F**) of rats treated with an IT 3 mg dose of ASO A with or without magnesium supplementation formulated in dACSF. Each point represents an individual animal. Data are expressed as mean ± s.d. The statistical significance was determined as follows: Kruskal-Wallis test *vs* vehicle (A, aCSF + magnesium; B, dACSF + magnesium) and Unpaired t-test (C-F). *P ≤ 0.05; ** P ≤ 0.01 *** P ≤ 0.001; **** P ≤ 0.0001. Abbreviations: IT= intrathecal; aCSF= artificial cerebrospinal fluid; dACSF= sodium deficient aCSF; aA, Acute neuronal activation.



Supplementary Figure 9. Dose-dependent ASO tissue concentration is observed with or without magnesium supplementation in male rats. (A-B). ASO tissue concentration plotted against increasing doses of ASO D in the cortex (A) and spinal cord (B) after 2 weeks of intrathecal administration with or without magnesium supplementation (40 mM) formulated in a modified aCSF (mACSF; magnesium and calcium removed) in male rats. Each point represents an individual animal (BW= 305 ± 9 g; n = 6). The specific number of animals per group and body weight are included in the supplementary table 19. Data are expressed as mean ± s.d. The significant difference was determined by the Unpaired t-test. *P ≤ 0.05; ** P ≤ 0.01 *** P ≤ 0.001; **** P ≤ 0.0001. No significant differences were found. Abbreviations: mACSF= modified artificial cerebrospinal fluid.

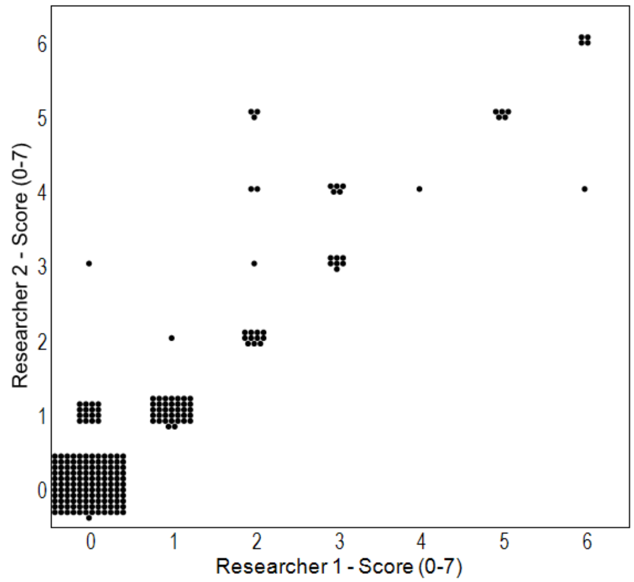


Figure supplementary 10. **Strong monotonic correlation of neuronal acute activation response score evaluated by two independent researchers.** Plot of acute activation response score of 3 different mouse studies evaluated blindly by 2 independent researchers. Each dot represents an individual evaluation. Spearman correlation statistical analysis was performed, P = 0.902, n = 228 evaluations.

Table supplementary 1. ASO sequences

ID	Simple sequence
ASO A	GCoAoCoCATATATATCTCoAoGAA
ASO B	AGoCoCoAATATTTATAGGoToGCT
ASO C	GToToGoGCCTTATCATACoCoTAA
ASO D (6)	GCoCoAoGoGCTGGTTATGAoCoTCA
ASO E	GToAoToCAGTCATTATCToCoATC
ASO F	AToToCoTCTTCATCTATCoAoCCC
ASO G	TToGoCoTTCACAACTCCCoAoCCT
ASO H	TGoToCoCCTAACATTTTCoCoTTA
ASO I	CToGoCoCAGACATATATCoAoTAA
ASO J	AAoToGoCTTTCATAACTToCoCAC
ASO K	GCoToCoATTTATTCTCAAoGoTAC
ASO L	TAoGoGoAGTTAACTAGAAoCoTCA
ASO M	CCoGoGoAAGTCTTCTCTAoAoGTC
ASO N	TCoToCoAGCTCCTTCCTAoGoGGC
ASO O	GCoAoAoATTGATTACCTCoAoGTC
ASO P	ACoToAoGCGGTTATCTTCoCoAAC
ASO Q	GToAoToGCTACATTCAAAoToCTA
siRNA A (35)	TUAoAoAoAoUoCoUoAoCoAoGoUoCoAoUoAoGoGoAAUAAUoUoUoUo AoGoAoUoGoUoCoAoGoUoAoUoCCU
siRNA B (36)	TGAoAoCoUoUoGoUoAoGoGoUoUoGoGoAoUoUoUoUCG AAAoAoUoCoCoAoAoCoCoUoAoCoAoAoGoUoUCA
Orange= 2'-O-methoxyethylribose; Black=DNA; PO= o; Yellow= 2'-O-MOE-(E)-5'-vinylphosphonate; Green= 2'-Fluororibose; Red= 2'-O-methylribose, Gray=2'-O Hexadecylribose**. *All linkages are PS unless noted**	

Table supplementary 2. Acute Inhibition scoring following IT administration in rodents

3h Acute Inhibition score	Rodent IT injection acute sedation scale
0	Bright, alert, responsive
1	No tone/movement in tail
2	Weak posterior posture
3	Hind limbs don't support weight, but can still move
4	Hind paws don't move – full hindlimb paralysis
5	Weak anterior posture
6	Fore paws don't move, but animal is still breathing
7	Death
Animals are observed 3 hours after dosing, and a score is assigned	

Table supplementary 3. Acute neuronal activation dose-response after ASO A intrathecal delivery in male rats (Figure 1A)

Treatment group	Number of rats (F/M)	BW grams (F/M)	ASO Conc. (mg/ml)	ASO (mM)	P value
aCSF	12M	285±9	0	0	
ASO A	11M	282±9	100	14.1	0-15min:0.000062 30-60min:<0.000001 90-120min: 0.000839
ASO A	12M	282±9	33.3	4.7	0-15min:0.040960 30-60min: 0.068323 90-120min: 0.089379
ASO A	12M	277±12	10	1.41	0-15min:0.003954 30-60min: 0.133050 90-120min: >0.999999
ASO A	12M	283±7	3.3	0.47	0-15min: 0.386558 30-60min: 0.017980 90-120min: >0.999999
ASO A	12M	281±10	1	0.3	0-15min:0.338110 30-60min: 0.450801 90-120min: >0.999999

All groups were dosed with 30µl. Abbreviations: aCSF= standard artificial cerebrospinal fluid. The Multiple Mann-Whitney tests determined the significant difference.

Table supplementary 4. Acute neuronal activation response after siRNA A and siRNA B intrathecal delivery in male rats (Figure 1B and Supplementary Figure 1A)

Treatment group	Number of rats (F/M)	BW grams (F/M)	siRNA Conc. (mg/ml)	siRNA (mM)	P value
aCSF	6M	271±7	0	0	
siRNA A	6M	291±6	100	6.7	0-15min:0.012987 30-60min: 0.012987 90-120min: 0.083333
siRNA A	6M	283±1	33.3	2.2	0-15min:0.058442 30-60min: 0.272727 90-120min: 0.454545
siRNA A	6M	280±2	10	0.67	0-15min:0.058442 30-60min: >0.999999 90-120min: 0.181818
siRNA A	6M	276±1	3.3	0.22	0-15min: 0.636364 30-60min:>0.999999 90-120min: 0.181818
siRNA A	6M	274±1	1	0.067	0-15min:0.740260 30-60min: 0.545455 90-120min: 0.060606
aCSF	4M	273±4	0	0	
siRNA B	6M	279±2	33.3	2.2	0-15min:0.066667 30-60min: 0.123810 90-120min: 0.571429
siRNA B	6M	284±1	10	0.67	0-15min:0.380952 30-60min: 0.533333 90-120min: 0.476190
siRNA B	6M	287±2	3.3	0.22	0-15min:0.111111 30-60min: 0.555556 90-120min: >0.999999
siRNA B	6M	292±1	1	0.067	0-15min: >0.999999 30-60min:>0.999999 90-120min: 0.380952

All groups were dose with 30µl. Abbreviations: aCSF= standard artificial cerebrospinal fluid. The significant difference was determined by the Multiple Mann-Whitney tests.

Table supplementary 5. Acute neuronal activation response after siRNA B intrathecal delivery in male rats (Supplementary Figure 1B)

Treatment group	Number of rats (F/M)	BW grams (F/M)	siRNA Conc. (mg/ml)	SiRNA (mM)	P value
PBS	4M	347±7	0	0	
siRNA B	4M	394±21	50	3.35	0-15min:0.0845
siRNA B	4M	410±23	100	6.7	0-15min:0.0845

All groups were dosed with 30μl. Abbreviations: aCSF= standard artificial cerebrospinal fluid. The Kruskal-Wallis test was used to determine the significant difference.

Table supplementary 6. Acute neuronal activation response after ASO A and ASO B intrathecal delivery in male rats (Supplementary Figure 1C)

Treatment group	Number of rats (F/M)	BW grams (F/M)	ASO Conc. (mg/ml)	ASO (mM)	P value vs aCSF	P value ASO A vs ASO B
aCSF	6M	284±5	0	0		
ASO A	6M	284±5	100	14.1	0-15min: 0.002165 30-60 min: 0.333333 90-120 min: 0.454545	0-15min:>0.999999 30-60 min: 0.707792 90-120 min: 0.727273
ASO B	6M	286±4	100	14	0-15min: 0.002165 30-60 min: 0.181818 90-120 min: >0.999999	

All groups were dosed with 3mg/ 30μl. Abbreviations: aCSF= standard artificial cerebrospinal fluid. The Multiple Mann-Whitney tests was used to determine the significant difference.

Table supplementary 7. Acute neuronal activation response after repeating intrathecal dosing of ASO B in male rats (Figure 1C-D)

Treatment group		Number of rats (F/M)	BW grams (F/M)			ASO Conc. (mg/ml)	ASO (mM)	P value
			Dose1	Dose 2	Dose 3			
Repeat dosing (group 1)	PBS	6M	149±10	370±35	401±39	0	0	
Repeat dosing (group 2)	ASO B	6M	141±9	360±12	398±21	100	14	0-15min: 0.0005 vs PBS 1st dose 0-15 min: 0.0167 vs PBS 2nd dose 0-15 min: 0.0526 vs PBS 3rd dose
Single dosing (group 3)	ASO B	3M	-	376±29	-	100	14	0-15min: >0.9999 Group 2 vs 3
Single dosing (group 4)	ASO B	3M	-	-	420±38	100	14	0-15min: >0.9999 Group 2 vs 4

All groups were dosed with 30μl. The Kruskal-Wallis test was used to determine the significant difference.

Table supplementary 8. Acute neuronal activation response after intrathecal delivery of ASO A in age-matched female and male rats (Figure 2A)

Treatment group	Number of rats (F/M)	BW grams (F/M)	ASO Conc. (mg/ml)	ASO (mM)	P value vs aCSF (females)	P value vs aCSF (males)
aCSF	4/4	204±4/ 260±5	0	0		
ASO A	4/4	202±6/ 261±8	100	14.1	0-15min: 0.0079 30-60min:0.0079 90-120min:0.0476	0-15min: 0.0286 30-60min:0.0286 90-120min:0.4286

All groups were dosed with 30µl. Abbreviations: aCSF= standard artificial cerebrospinal fluid. The Mann-Whitney test was used to determine the significant difference.

Table supplementary 9. Acute neuronal activation response after intrathecal delivery of 16 ASOs, same chemistry, different sequences in male rats (Figure 1F)

Treatment group	Number of rats (F/M)	BW grams (F/M)	ASO Conc. (mg/ml)	ASO (mM)	P value vs ASO A
PBS	4M	290±14	0	0	0-15min:0.0281
ASO A	4M	292±13	100	14.1	
ASO B	4M	292±9	100	14.1	0-15min: >0.9999
ASO C	4M	290±11	100	14.1	0-15min: >0.9999
ASO E	4M	291±6	100	14.1	0-15min: >0.9999
ASO F	4M	291±8	100	14.1	0-15min: 0.5086
ASO G	4M	288±7	0	0	0-15 min: >0.9999
ASO H	4M	285±12	100	14.1	0-15min: >0.9999
ASO I	4M	292±5	100	14.1	0-15min: 0.1747
ASO J	4M	292±6	100	14.1	0-15min: >0.9999
ASO K	4M	299±17	100	14.1	0-15min: >0.9999
ASO L	4M	296±8	100	14.1	0-15min: >0.9999
ASO M	4M	297±9	100	14.1	0-15min: >0.9999
ASO N	4M	299±7	100	14.1	0-15min: >0.9999
ASO O	4M	292±7	100	14.1	0-15min: >0.9999
ASO P	4M	292±9	100	14.1	0-15min: >0.9999
ASP Q	4M	296±8	100	14.1	0-15min: >0.9999

All groups were dosed with 30µl. The Kruskal-Wallis test was used to determine the significant difference.

Table supplementary 10. Acute neuronal activation dose response after ICV delivery of ASO A in mice (Figure 2B)

Treatment group	Number of mice (F/M)	BW grams (F/M)	ASO Conc. (mg/ml)	ASO (mM)	P value
aCSF	4/4	18±0.5/ 21±0.5	0	0	
ASO A	4/4	17±0.7/ 21±1	30	4.2	0-15min:>0.999999 30-60min:0.466667 90-120min: >0.999999
ASO A	4/4	18±1/ 21±1	50	7.1	0-15min:>0.999999 30-60min: 0.076923 90-120min: >0.999999
ASO A	4/4	17±1/ 23±2	70	9.9	0-15min:0.006993 30-60min: 0.076923 90-120min: 0.466667
ASO A	4/4	18±1/ 22±1	100	14.1	0-15min: 0.001399 30-60min: 0.001399 90-120min: >0.999999

All groups were dosed with 10µl. Abbreviations: aCSF= standard artificial cerebrospinal fluid. The Multiple Mann-Whitney tests determined the significant difference vs aCSF

Table supplementary 11. Acute neuronal activation response after ICV delivery of ASO A in female and male mice with high and low body weight (Figure 2C-D)

Treatment group	Number of mice (F/M)	BW grams (F/M)	ASO Conc. (mg/ml)	ASO (mM)	P value vs aCSF (females)	P value vs aCSF (males)
aCSF	5/5	17±0.4/ 18±0.5	0	0		
ASO A	5/7	17±0.3/ 17±1	100	14.1	0-15min: 0.0079 30-60min:0.0079 90-120min:0.1667	0-15min: 0.0025 30-60min:0.0025 90-120min:>0.9999
aCSF	8/4	23±1/ 22±2	0	0		
ASO A	8/8	23±1/ 23±2	100	14.1	0-15min:0.5096 30-60min:0.0631 90-120min:>0.9999	0-15min: 0.5152 30-60min:0.0040 90-120min:>0.9999

All groups were dosed with 1000 µg/10µl. Abbreviations: aCSF= standard artificial cerebrospinal fluid. The Mann-Whitney test was used to determine the significant difference.

Table supplementary 12. Formulation description of ASO A with or without increasing concentrations of magnesium supplementation in rats (Figure 3A, Supplementary Figures 4A-B and 6A-B)

Treatment group	**n of rats (F/M)**	**BW (F/M)**	**ASO Conc. (mg/ml)**	**ASO (mM)**	**Ca^{++} (mM)**	**Mg^{++} (mM)**	**$MgCl_2$ supplemented (mM)**	**DivCat: ASO ratio**	**P value vs aCSF group**
aCSF	4/4	231±14/ 283±4	0	0	1.4	0.8	0	-	-
aCSF+ 42.2 mM total DivCat (Mg supp)	4/4	226±13/ 285±10	0	0	1.4	0.8	40	**0**	0-15min: >0.9999 30-60 min:>0.9999 90-120 min:>0.9999
aCSF+ ASO A	4/4	224±4/ 289±6	100	14.1	1.4	0.8	0	**0.16:1**	0-15min: 0.0002 30-60 min: 0.0028 90-120 min: 0.9165
aCSF+ 4.2 mM total DivCat (Mg supp)+ ASO A	4/4	223±7/ 290±14	100	14.1	1.4	0.8	2	**0.3:1**	0-15min: 0.0002 30-60 min:0.0013 90-120 min:0.0316
aCSF+ 10 mM total DivCat (Mg supp)+ ASO A	4/4	226±8/ 290±10	100	14.1	1.4	0.8	7.8	**0.7:1**	0-15min: 0.4667 30-60 min: 0.0640 90-120 min:>0.9999
aCSF+ 18.5 mM total DivCat (Mg supp)+ ASO A	4/4	226±10/ 288±8	100	14.1	1.4	0.8	16.3	**1.3:1**	0-15min: 0.2465 30-60 min: 0.1003 90-120 min:>0.9999
aCSF+ 24 mM total DivCat (Mg supp)+ ASO A	4/4	224±7/ 282±8	100	14.1	1.4	0.8	21.8	**1.7:1**	0-15min: >0.9999 30-60 min: 0.9194 90-120 min:0.1638
aCSF+ 32.5 mM total DivCat (Mg supp)+ ASO A	4/4	224±4/ 286±3.2	100	14.1	1.4	0.8	30.3	**2.3:1**	0-15min: >0.9999 30-60 min:>0.9999 90-120 min:>0.9999
aCSF+ 42.2 mM total DivCat (Mg supp)+ ASO A	4/4	225±4/ 286±7	100	14.1	1.4	0.8	40	**3:1**	0-15min: >0.9999 30-60 min:>0.9999 90-120 min:>0.9999

All groups were dosed at 30 mg/ 30µl. Abbreviations: DivCat= divalent cations; Mg supp= magnesium supplemented; aCSF= standard artificial cerebrospinal fluid. The significant difference was determined by the Kruskal-Wallis test.

Table supplementary 13. Formulation description of ASO A with or without increasing concentrations of calcium supplementation in rats (Figure 3B, Supplementary Figures 4C-D and 6C-D)

Treatment group	n of rats (F/M)	BW (F/M)	ASO Conc. (mg/ml)	ASO (mM)	Ca^{++} (mM)	Mg^{++} (mM)	$MgCl_2$ supplemented (mM)	DivCat: ASO ratio	P value vs aCSF group
aCSF	4/4	226±16/ 300±15	**0**	**0**	**1.4**	**0.8**	**0**	**0**	-
aCSF+ 42.2 mM total DivCat (Ca supp)	4/4	229±9/ 298±12	0	0	1.4	0.8	40	**0**	0-15min: >0.9999 30-60 min:>0.9999 90-120 min: >0.9999
aCSF+ ASO A	4/4	223±/11 298±5	100	14.1	1.4	0.8	0	**0.16:1**	0-15min: <0.0001 30-60 min: 0.1346 90-120 min: >0.9999
aCSF+ 4.2 mM total DivCat (Ca supp)+ ASO A	4/4	222±10/ 290±10	100	14.1	1.4	0.8	2	**0.3:1**	0-15min: 0.0353 30-60 min: 0.1633 90-120 min: 0.2186
aCSF+ 10 mM total DivCat (Ca supp)+ ASO A	4/4	222±8/ 285±10	100	14.1	1.4	0.8	7.8	**0.7:1**	0-15min:>0.9999 30-60 min: 0.2608 90-120 min: 0.0260
aCSF+ 18.5 mM total DivCat (Ca supp)+ ASO A	4/4	224±9/ 277±14	100	14.1	1.4	0.8	16.3	**1.3:1**	0-15min: >0.9999 30-60 min: >0.9999 90-120 min: 0.0069
aCSF+ 24 mM total DivCat (Ca supp)+ ASO A	4/4	222±7/ 286±1	100	14.1	1.4	0.8	21.8	**1.7:1**	0-15min: >0.9999 30-60 min: 0.0004 90-120 min: 0.1615
aCSF+ 32.5 mM total DivCat (Ca supp)+ ASO A	4/4	226±9/ 287±3	100	14.1	1.4	0.8	30.3	**2.3:1**	0-15min: >0.9999 30-60 min: 0.0059 90-120 min:<0.0001
aCSF+ 42.2 mM total DivCat (Ca supp)+ ASO A	4/4	224±7/ 288±4	100	14.1	1.4	0.8	40	**3:1**	0-15min:>0.9999 30-60 min: 0.0002 90-120 min: 0.0030

All groups were dosed at 30 mg/ 30µl. Abbreviations: DivCat= divalent cations; Ca supp= calcium supplemented; aCSF= standard artificial cerebrospinal fluid. The significant difference was determined by the Kruskal-Wallis test.

Table supplementary 14. Formulation description of ASO A with or without increasing concentrations of calcium + magnesium supplementation in rats (Figure 3C, Supplementary Figures 4E-F and 6E-F)

Treatment group	n of rats (F/M)	BW (F/M)	ASO Conc. (mg/ml)	ASO (mM)	Ca^{++} (mM)	Mg^{++} (mM)	$MgCl_2$ suppl (mM)	$CaCl_2$ supp (mM)	DivCat: ASO ratio	P value vs aCSF group
aCSF	4/4	224±12/ 300±12	**0**	**0**	**1.4**	**0.8**	0	0	**0**	-
aCSF+42.2 mM total DivCat (Mg+Ca supp)	4/4	224±/10 296±4	0	0	1.4	0.8	25.5	14.6	**0**	0-15min: >0.9999 30-60 min:>0.9999 90-120 min: >0.9999
aCSF+ASO A	4/4	226±/4 297±9	100	14.1	1.4	0.8	0	0	**0.16:1**	0-15min: 0.0002 30-60 min: 0.0119 90-120 min: >0.9999
aCSF+4.2 mM total DivCat (Mg+Ca supp)+ASO A	4/4	224±11/ 296±5	100	14.1	1.4	0.8	1.29	0.74	**0.3:1**	0-15min: 0.0524 30-60 min: 0.0767 90-120 min: >0.9999
aCSF+10 mM total DivCat (Mg+Ca supp)+ASO A	4/4	226±4/ 291±10	100	14.1	1.4	0.8	4.96	2.84	**0.7:1**	0-15min:>0.9999 30-60 min: >0.9999 90-120 min: 0.1274
aCSF+18.5 mM total DivCat (Mg+Ca supp)+ASO A	4/4	221±5/ 291±9	100	14.1	1.4	0.8	10.4	5.9	**1.3:1**	0-15min:>0.9999 30-60 min:>0.9999 90-120 min: 0.4049
aCSF+24 mM total DivCat (Mg+Ca supp)+ASO A	4/4	226±1/ 296±5	100	14.1	1.4	0.8	15.8	9.0	**1.7:1**	0-15min:>0.9999 30-60 min: 0.0001 90-120 min: 0.0002
aCSF+32.5 mM total DivCat (Mg+Ca supp)+ASO A	4/4	225±11/ 293±6	100	14.1	1.4	0.8	19.3	11.0	**2.3:1**	0-15min:>0.9999 30-60 min: 0.0005 90-120 min: <0.0001
aCSF+42.2 mM total DivCat (Mg+Ca supp)+ASO A	4/4	221±11/ 294±4	100	14.1	1.4	0.8	25.5	14.6	**3:1**	0-15min:>0.9999 30-60 min: 0.0087 90-120 min: 0.0108

All groups were dosed at 30 mg/ 30µl. Abbreviations: DivCat= divalent cations; Mg+Ca supp= Magnesium plus calcium supplemented; aCSF= standard artificial cerebrospinal fluid. The significant difference was determined by the Kruskal-Wallis test.

Table supplementary 15. Formulation description of ASO A with or without increasing concentrations of calcium supplementation formulated in dACSF (isotonic) in rats (Supplementary Figure 5)

Treatment group	n of rats (F/M)	BW (F/M)	ASO Conc. (mg/ ml)	ASO (mM)	Ca^{++} (mM)	Mg^{++} (mM)	$MgCl_2$ suppl (mM)	$NaCl_2$ (mM)	Tonicity (mOsm/ ml)	DivCat: ASO ratio	P value vs aCSF group
aCSF	4/4	228±8/ 295±30	**0**	**0**	**1.4**	**0.8**	**0**	**150**	**292**	**0**	-
dACSF+ ASO A	4/4	227±9/ 307±14	100	14.1	1.4	0.8	0	42	**286**	**0**	0-15min: 0.0006 30-60 min: 0.0755 90-120 min: 0.0596
dACSF+7 mM total DivCat (Ca supp)+ASO A	4/4	224±7/ 298±14	100	14.1	1.4	0.8	4.8	42	**304**	**0.5:1**	0-15min: :>0.9999 30-60 min: 0.0024 90-120 min: 0.0004
dACSF+21 mM total DivCat (Ca supp)+ASO A	4/4	224±7/ 299±8	100	14.1	1.4	0.8	18.8	0	**260**	**1.5:1**	0-15min: 0.1264 30-60 min: 0.0037 90-120 min: 0.0064
dACSF+42.2 mM total DivCat (Ca supp)+ASO A	4/4	224±5/ 301±11	100	14.1	1.4	0.8	40	0	**311**	**3:1**	0-15min:0.1145 30-60 min: 0.0003 90-120 min: 0.0256

All groups were dosed at 30 mg/ 30µl. Abbreviations: DivCat= divalent cations; Ca supp= calcium supplemented; $MgCl_2$ suppl= magnesium supplemented; aCSF= standard artificial cerebrospinal fluid; dACSF= sodium deficient aCSF. The significant difference is determined by the Kruskal-Wallis test.

Table supplementary 16. Formulation description of ASO C with or without increasing concentrations of magnesium supplementation in male rats (Supplementary Figure 7)

Treatment group	n of rats (F/M)	BW (F/M)	ASO Conc. (mg/ml)	ASO (mM)	Ca^{++} (mM)	Mg^{++} (mM)	$MgCl_2$ supplemented (mM)	DivCat: ASO ratio	P value vs aCSF+Mg
aCSF+ 32.2 mM total DivCat (Mg supp)	4M	259±6	0	0	1.4	0.8	30	**0**	-
aCSF+ ASO C	4M	264±16	100	14.1	1.4	0.8	0	**0.16:1**	0-15min: 0.0380
aCSF+ 3.2 mM total DivCat (Mg supp)+ ASO C	4M	268±2	100	14.1	1.4	0.8	1	**0.23:1**	0-15min: 0.4994
aCSF+ 5.2 mM total DivCat (Mg supp)+ ASO C	4M	261±2	100	14.1	1.4	0.8	3	**0.37:1**	0-15min: 0.1508
aCSF+ 12.2 mM total DivCat (Mg supp)+ ASO C	4M	257±2	100	14.1	1.4	0.8	10	**0.86:1**	0-15min: >0.9999
aCSF+ 32.2 mM total DivCat (Mg supp)+ ASO C	4M	267±7	100	14.1	1.4	0.8	30	**2.3:1**	0-15min: >0.9999

All groups were dosed at 30 mg/ 30µl. Abbreviations: DivCat= divalent cations; Mg supp= magnesium supplemented; aCSF= standard artificial cerebrospinal fluid. The significant difference was determined by the Kruskal-Wallis test.

Table supplementary 17. Formulation description of ASO A with or without magnesium supplementation in hypertonic or tonicity-controlled conditions in rats (Figure 4, Supplementary Figure 8)

Treatment group	n of rats (F/M)	BW (F/M)	ASO Conc. (mg/ml)	ASO (mM)	Ca^{++} (mM)	Mg^{++} (mM)	$MgCl_2$ suppl (mM)	$NaCl_2$ (mM)	Tonicity (mOsm/ml)	DivCat: ASO ratio	P value vs vehicle + Mg supp group
aCSF	4/4	204±4/ 261±5	**0**	**0**	**1.4**	**0.8**	**0**	**150**	**293**	0	0-15min: >0.9999 30-60 min: >0.9999 90-120 min: >0.9999
aCSF+ 42.2 mM total DivCat (Mg supp)	4/4	205±6/ 260±5	0	0	1.4	0.8	40	150	**400**	0	-
aCSF+ ASO A	5/4	202±6/ 260±8	100	14.1	1.4	0.8	0	150	**492**	0	0-15min:<0.0001 30-60 min: <0.0001 90-120 min: 0.0276
aCSF+ 42.2 mM total DivCat (Mg supp) + ASO A	5/5	206±2/ 263±5	100	14.1	1.4	0.8	40	150	**616**	3:1	0-15min: >0.9999 30-60 min: 0.7689 90-120 min: 0.0027
dACSF+ 42.2 mM total DivCat (Mg supp)	4/4	198±8/ 270±11	0	0	1.4	0.8	40	96	**311**	0	-
dACSF+ ASO A	6/5	204±9/ 268±13	100	14.1	1.4	0.8	0	42	**285**	0	0-15min:<0.0001 30-60 min: <0.0001 90-120 min: 0.0833
dACSF+ 42.2 mM total DivCat (Mg supp) + ASO A	6/6	201±16/ 278±5	100	14.1	1.4	0.8	40	0	**325**	3:1	0-15min:>0.9999 30-60 min: 0.7248 90-120 min: >0.9999

All groups were dosed at 30 mg/ 30µl. Abbreviations: DivCat= divalent cations; $MgCl_2$ suppl= magnesium supplemented; aCSF= standard artificial cerebrospinal fluid; dACSF= sodium deficient aCSF. The significant difference was determined by the Kruskal-Wallis test.

Table supplementary 18. Formulation description of ASO A with or without magnesium supplementation formulated in mACSF and aCSF in transgenic *ATXN3* mice (Figure 5A-F)

Treatment group	n of mice (F/M)	BW grams (F/M)	ASO Conc. (mg/ml)	ASO (mM)	Ca^{++} (mM)	Mg^{++} (mM)	$MgCl_2$ supp (mM)	ASO dose (µg)	P value vs vehicle ATXN3 mRNA CX/SC	P value ASO conc CX/SC
aCSF	2/2	19±1/ 24±1	0	0	1.4	0.8	0	223	-	-
aCSF+ 40 mM total $MgCl_2$	2/2	19±0.7/ 23±1	0	0	1.4	0.8	39.2	0	0.9985/0.9955	-
aCSF+ ASO 22.3 mg	3/3	18±1/ 22±1	22.3	3.14	1.4	0.8	0	223	<0.0001/<0.0001	
aCSF+ 40 mM total $MgCl_2$ + ASO 22.3 mg	3/3	19±0.7/ 24±0.9	22.3	3.14	1.4	0.8	0	223	<0.0001/<0.0001	>0.9999/0.4067
mACSF+ 40 mM total $MgCl_2$	2/2	20±0.4/ 21±0.5	0	0	0	0	40	0	>0.9999/0.9362	-
mACSF+ ASO 22.3 mg	2/3	18±2/ 22±0.5	22.3	3.14	0	0	0	223	<0.0001/<0.0001	0.9904/0.7095
mACSF+ 40 mM total $MgCl_2$ + ASO 22.3 mg	3/3	19±0.8/ 22±0.5	22.3	3.14	0	0	40	223	<0.0001/<0.0001	0.9883/0.9871

All mice received a dose volume of 10 µl. Abbreviations: DivCat= divalent cations; $MgCl_2$= magnesium chloride; aCSF= standard artificial cerebrospinal fluid; mACSF= modified artificial cerebrospinal fluid. The significance difference was determined by one-way ANOVA followed by Tukey's post hoc test.

Table supplementary 19. Formulation description of ASO D with or without $MgCl_2$ supplementation formulated in modified artificial cerebrospinal fluid (mACSF) in male rats (Figure 5G-L, Supplementary Figure 9)

Treatment group	n of rats (F/M)	BW (F/M)	ASO Conc. (mg/ml)	ASO (mM)	Ca^{++} (mM)	Mg^{++} (mM)	$MgCl_2$ supp (mM)	ASO dose (mg)	P value ASO Conc. CX/SC
mACSF	6M	300±5	0	0	0	0	0	**0**	-
mACSF+ 40 mM total $MgCl_2$	6M	314±8	0	0	0	0	40	**0**	-
mACSF+ ASO	6M	298±9	1	0.14	0	0	0	**0.03**	-
mACSF+ ASO	6M	298±4	3.3	0.46	0	0	0	**0.1**	-
mACSF+ ASO	6M	299±7	10	1.4	0	0	0	**0.3**	-
mACSF+ ASO	6M	303±8	33	4.6	0	0	0	**1**	-
mACSF+ 40 mM total $MgCl_2$ + ASO	6M	314±9	1	0.14	0	0	40	**0.03**	0.491146/ 0.810752
mACSF+ 40 mM total $MgCl_2$ + ASO	6M	309±11	3.3	0.46	0	0	40	**0.1**	0.515706/ 0.069634
mACSF+ 40 mM total $MgCl_2$ + ASO	6M	310±10	10	1.4	0	0	40	**0.3**	0.038847/ 0.771993
mACSF+ 40 mM total $MgCl_2$ + ASO	6M	304±5	33	4.6	0	0	40	**1**	0.729639/ 0.492048

All groups received a dose volume of 30µl. Abbreviations: DivCat= divalent cations; MgCl2= magnesium chloride; mACSF= modified artificial cerebrospinal fluid. . The significant difference was determined by the Unpaired t-test groups without Mg supp vs Groups with Mg supp

Table supplementary 20. Formulation description of ASO A dosed at different concentrations and volumes with or without magnesium supplementation in rats (Figure 6)

	Treatment group	n of rats (F/M)	BW (F/M)	ASO dose (mg)	Dose Volume (µl)	ASO Conc. (mg/ ml)	ASO (mM	Ca^{++} (mM)	Mg^{++} (mM)	$MgCl_2$ suppl (mM)	DivCat : ASO ratio	P value
A	aCSF+ 42.2 mM total DivCat (Mg supp)	4/4	215±13/ 288±10	**0**	**30**	**0**	**0**	**1.4**	0.8	**40**	-	-
B	aCSF+ ASO A	4/4	218±11/ 288±11	3	30	100	14.1	1.4	0.8	0	0.16:1	<0.0001 *vs* A
C	aCSF+42.2 mM total DivCat (Mg supp)+ ASO A	4/4	220±9/ 290±5	3	30	100	14.1	1.4	0.8	40	3:1	0.0003 *vs* B
D	aCSF+ 42.2 mM total DivCat (Mg supp)	4/4	214±9/ 288±11	0	100	0	0	1.4	0.8	40	-	-
E	aCSF+ ASO A	4/4	217±7/ 285±6	3	100	30	4.22	1.4	0.8	0	0.52:1	0.0017 *vs* D
F	aCSF+ 12.7 mM total DivCat (Mg supp)+ ASO A	4/4	219±8/ 284±4	3	100	30	4.22	1.4	0.8	10.5	3:1	0.0399 *vs* E
G	aCSF+ ASO C	4M	271±13	3	30	100	14.1	1.4	0.8	0	0.16:1	-
H	aCSF+ ASO C	4M	270±13	3	100	30	4.22	1.4	0.8	0	0.52:1	0.8088 *vs* G
I	aCSF+ 12.2 mM total DivCat (Mg supp)+ ASO C	4M	251±14	3	100	30	4.22	1.4	0.8	10	2.9:1	0.0149 *vs* G 0.2639 *vs* H

Abbreviations: DivCat= divalent cations; Mg supp= magnesium supplemented; aCSF= standard artificial cerebrospinal fluid. The significant difference was determined by the Kruskal-Wallis test.

Table supplementary 21. Detailed description of ASO A formulations used in Non-human primates' experiments (Figure 7)

Treatment group NHP	n of NHP (F/M)	BW, Kg (F/M)	ASO conc. (mg/ml)	ASO (mM)	Ca++ (mM)	Mg++ (mM)	$MgCl_2$ supp (mM)	Total DivCat (mM)	DivCat: ASO ratio	aA score (average)	P value
aCSF + 23.2 mM total DivCat (Mg supp)	2/2	2.7±1/ 3.2±0.5	-	-	1.4	0.8	21	23.2	**-**	1	
ASO 70 mg	3/3	2.7±0.1/ 3.2±0.4	70	9.9	1.4	0.8	0	2.2	**0.2**	3.25	0.0348 vs aCSF + 9.2 mM DivCat
9.2 mM total DivCat (Mg supp) + ASO 70 mg	3/3	2.8±0.2/ 3.3±0.9	70	9.9	1.4	0.8	7	9.2	**0.9**	2	
23.2 mM total DivCat (Mg supp) + ASO 70 mg	2/2	2.7±0.2/ 3.3±0.6	70	9.9	1.4	0.8	21	23.2	**2.3**	0.87	0.0231 vs aCSF + ASO
23.2 mM total DivCat (Mg supp) + ASO 105 mg	3/3	2.7±0.1/ 3.3±0.4	105	14.8	1.4	0.8	21	23.2	**1.6**	3.5	
23.2 mM total DivCat (Mg supp) + ASO 140 mg	3/3	2.7±0.2/ 3.3±0.8	140	19.7	1.4	0.8	21	23.2	**1.2**	4	0.0306 vs ASO + 23.2 mM Div Cat
aCSF+ 33.2 mM total DivCat (Mg supp) + ASO 105 mg	2/2	2.7±0.1/ 3.3±0.5	105	14.8	1.4	0.8	31	33.2	**2.24**	1.25	>0.9999 vs 44.5 mM DivCat + ASO 140
44.5 mM total DivCat (Mg supp) + ASO 140 mg	3/3	2.7±0.1/ 3.4±0.4	140	19.7	1.4	0.8	42.3	44.5	**2.25**	2	
58.3 mM total DivCat (Mg supp) + ASO 140 mg	3/3	2.8±0.3/ 3.3±0.8	140	19.7	1.4	56.9	56.1	58.3	**2.95**	1	>0.9999 vs 33.2 mM DivCat + ASO 105

The volume injected in all groups = 1 ml. Abbreviations: DivCat=divalent cations; supp=supplemented; Aa score= acute activation score. The significant difference was determined by the Kruskal-Wallis test.

Table supplementary 22. Primer sequences used in the study

Target	Species	Primers	
Malat1	Rat	Forward	CGTTAGACTTTTGTCACCTCACTTGA
		Reverse	GGCAAGCGCTTATATGCAATC
		Probe	TTTGCAGAGGCCTCATTTCATCCTTCA
Gfap	Rat	Forward	GAGAGAGATTCGCACTCAGTACGA
		Reverse	GTCTGCAAACTTGGACCGATACCA
		Probe	CAGTGGCCACCAGTAACATGCAAGAAAC
Aif1	Rat	Forward	AGGAGAAAAACAAAGAACACCAGAA
		Reverse	CAATTAGGGCAACTCAGAAATAGCT
		Probe	CCAACTGGTCCCCCAGCCAAGAX
ATXN3	Human	Forward	TGACACAGACATCAGGTACAAATC
		Reverse	TGCTGCTGTTGCTGCTT
		Probe	AGCTTCGGAAGAGACGAGAAGCCTA
Gapdh	Mouse	Forward	GGCAAATTCAACGGCACAGT
		Reverse	GGGTCTCGCTCCTGGAAGAT
		Probe	AAGGCCGAGAATGGGAAGCTTGTCATC
Gapdh	Rat	Forward	TGCTCCTCCCTGTTCTAGAGACA
		Reverse	CACCGACCTTCACCATCTTGT
		Probe	CCGCATCTTCTTGTGCAGTGCCAG

Table supplementary 23. Equilibrium dialysis of different concentrations of ASO A or SiRNA A against buffer (Sodium acetate or aCSF) with $CaCl_2$ and $MgCl_2$ (Figure 8)

aCSF buffer		
ASO A mM	**Mg^{2+} mM**	**Ca^{2+} mM**
0.25	0.3	0.4
0.5	0.5	0.6
1	0.8	1.3
2.5	1.6	2.8
5	3.1	5.3
10	6.1	10.0
15	7.9	13.5
20	9.3	15.0

Mg^{2+} and Ca^{2+} concentrations above aCSF basal levels

aCSF buffer		
siRNA A mM	**Mg^{2+} mM**	**Ca^{2+} mM**
0.25	1.3	1.4
0.5	2.2	2.8
1	4.0	6.2
2.5	9.2	14.1
5	14.2	26.3
10	24.9	46.9
15	38.3	68.3

Mg^{2+} and Ca^{2+} concentrations above aCSF basal levels

Sodium acetate buffer			
Mg^{2+} mM	**Bound Mg^{2+} mM**	**Ca^{2+} mM**	**Bound Ca^{2+} mM**
0	0	**0**	0
2	5.8	**2**	7.5
5	10.6	**5**	14.4
10	17	**10**	19.9
40	29.8	**40**	28.8
80	55.2	**80**	41.3
200	75.9	**200**	49.3

10 mM Sodium acetate and 150 mM sodium chloride, pH 7.2.